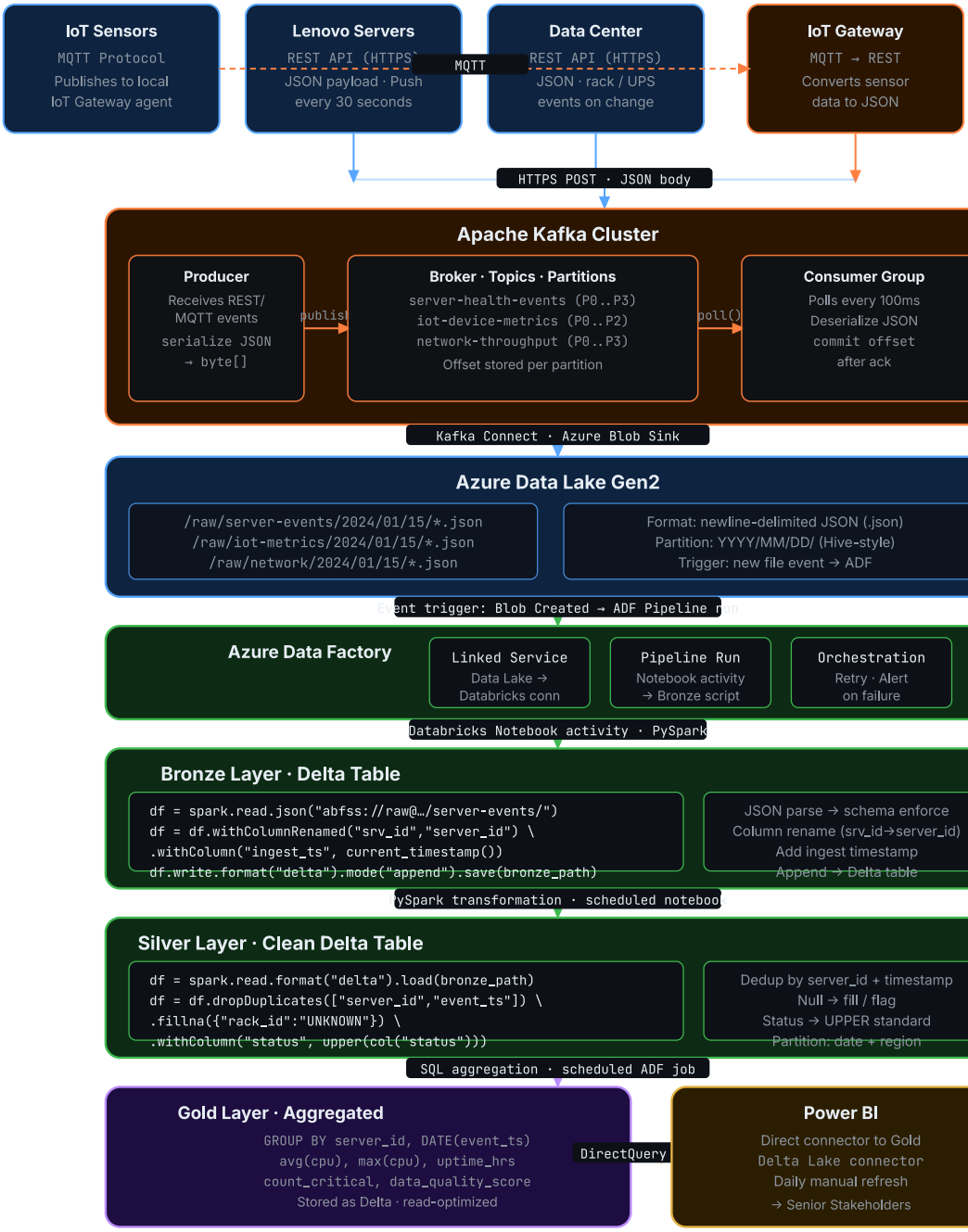
LENOVO · AZURE STACK · TECHNICAL DEEP DIVE

How Each Component Connects

Protocol-level connections — what talks to what, and how



KAFKA CONNECTION

- → Producer serializes JSON → byte array
- → Topic partitioned by server_id hash
- → Consumer polls every 100ms
- → Offset committed after successful write

DATA LAKE CONNECTION

- → abfss:// protocol (Azure Blob FS secure)
- → Hive-style partition: YYYY/MM/DD/
- → New file event triggers ADF pipeline
- → Newline-delimited JSON format

- → Kafka Connect → Azure Blob Sink connector
- → Storage Account Key auth

ADF → DATABRICKS

- → Linked Service: Databricks workspace URL
- → Pipeline: Notebook Activity type
- → Passes notebook path + parameters
- → Waits for cluster to start (auto-terminate)
- → Retry 3x on failure · email alert

POWER BI CONNECTION

- → Azure Databricks connector in Power BI
- → DirectQuery mode → Gold Delta table
- → Manual refresh daily by Data Engineer
- → Published to Power BI workspace
- → Shared with senior stakeholders